

HealthConnect Week 4 Initial Analysis Document

1. Dataset Overview

Project Background

HealthConnect needs to better understand patient appointment attendance and no-show patterns. Missed appointments can affect clinic operations, staff scheduling, service delivery, and the efficient use of available appointment slots.

The focus of the data analytics track is to explore the appointment dataset and establish a foundation for investigating the factors associated with appointment attendance, no-shows, and cancellations.

The analysis will use SQL and Power BI. SQL will support data exploration, validation, cleaning, and analysis, while Power BI will be used in subsequent stages to analyse trends and communicate insights through visualisations and dashboards.

Dataset Structure

The HealthConnect appointment dataset contains 5,000 records and 18 variables. Each record represents an appointment.

The variables available in the dataset are:

Variable	Description
appointment_id	Unique identifier for each appointment
patient_id	Unique identifier for each patient
gender	Patient gender
age	Patient age
age_group	Age category of the patient
appointment_type	Type of appointment booked
booking_date	Date the appointment was booked
appointment_date	Scheduled date of the appointment
appointment_day	Day of the week of the appointment
appointment_time	Time category of the appointment
booking_lead_days	Number of days between booking and appointment
previous_appointments	Number of previous appointments for the patient
previous_no_shows	Number of previous appointments missed by the patient
reminder_sent	Indicates whether an appointment reminder was sent
reminder_channel	Channel used to send the reminder
distance_to_clinic_km	Estimated distance between the patient and the clinic
waiting_time_minutes	Waiting time associated with the appointment
appointment_outcome	Final appointment outcome, including Attended, No-Show, or Cancelled

The main outcome variable for this project is `appointment_outcome`. This variable will be used to investigate attendance and no-show patterns.

The dataset currently contains the following appointment outcomes:

- No-Show: 2,423 records
- Attended: 2,314 records
- Cancelled: 263 records

This shows that appointment attendance and no-shows are both significant areas for investigation.

2. Initial Data Quality Assessment

An initial review of the dataset was conducted to identify potential data quality issues before further analysis.

Missing Values

Most variables contain complete data. However, missing values were identified in the following columns:

- reminder_channel: 1,366 missing values
- distance_to_clinic_km: 90 missing values
- waiting_time_minutes: 60 missing values

The missing values in reminder_channel appear to be logically related to the reminder_sent variable. Where a reminder was not sent, no reminder channel was recorded. Therefore, these missing values should not automatically be treated as errors. During analysis, they may be interpreted as "No Reminder Sent" where appropriate.

The missing values in distance_to_clinic_km and waiting_time_minutes will require further consideration. Their treatment will depend on the purpose of the analysis and whether the missing records could affect the interpretation of no-show patterns.

Duplicate Records

No fully duplicated records were identified in the dataset.

Further analysis should still confirm that appointment_id is unique, as it is expected to serve as the unique identifier for each appointment.

Data Types

The initial dataset structure shows that:

- Booking and appointment dates are stored as date values.
- Age, booking lead days, previous appointments, and previous no-shows are stored as numeric values.
- Distance and waiting time are stored as decimal numeric values.
- Gender, appointment type, appointment day, appointment time, reminder information, and appointment outcome are categorical variables.

The data types are generally appropriate for the planned analysis.

Initial Validity and Consistency Checks

Several logical checks were performed on the dataset:

- Age ranges from 18 to 80 years.
- Booking lead days range from 0 to 60 days.
- Previous appointments range from 0 to 11.
- Previous no-shows range from 0 to 5.
- Distance to clinic ranges from 0.5 km to 45 km.
- Waiting time ranges from 2 to 68 minutes.
- No cases were identified where previous no-shows exceeded previous appointments.
- Age values are consistent with the recorded age groups.
- Appointment dates are not earlier than booking dates.
- Appointment day values are consistent with the actual day of the week derived from the appointment date.

- Reminder channels are recorded when reminders were sent and are blank when reminders were not sent.

Based on this initial assessment, the dataset appears generally suitable for analysis. However, missing values and the treatment of cancellations will need to be considered carefully during the next stages.

3. Important Variables Relevant to Appointment Attendance and No-Shows

The following variables are considered important for investigating appointment attendance patterns.

appointment_outcome

This is the primary outcome variable. It identifies whether an appointment was attended, resulted in a no-show, or was cancelled. It will be central to measuring and analysing attendance patterns.

booking_lead_days

This variable shows how far in advance an appointment was booked. It may help determine whether appointments booked further in advance are associated with different no-show patterns.

previous_no_shows

This variable provides information about a patient's previous history of missed appointments. It may help identify whether patients with previous no-shows are more likely to miss future appointments.

previous_appointments

This provides context about the patient's appointment history and level of previous engagement with HealthConnect.

reminder_sent

This variable can help investigate whether sending reminders is associated with improved appointment attendance.

reminder_channel

Where reminders were sent, the reminder channel may help determine whether attendance patterns differ between SMS, WhatsApp, and email reminders.

appointment_type

Different types of appointments may have different attendance and no-show patterns. For example, patients may behave differently when attending follow-up appointments compared with specialist consultations or diagnostic tests.

appointment_date and appointment_day

These variables can be used to investigate whether attendance patterns vary over time or by day of the week.

appointment_time

Morning, afternoon, and evening appointments may experience different attendance patterns.

age and age_group

These variables can be used to investigate whether attendance and no-show behaviour differ across patient age categories.

distance_to_clinic_km

Distance may be associated with a patient's ability or willingness to attend an appointment. Further analysis can investigate whether patients living farther from the clinic have higher no-show rates.

waiting_time_minutes

This variable may provide insight into the patient experience and can be explored to determine whether waiting time is associated with attendance patterns.

4. Relevant Business Questions

The following business questions will guide the subsequent stages of the HealthConnect analysis:

- What proportion of scheduled appointments result in attendance, no-shows, and cancellations?

This will establish the overall scale of the appointment attendance problem.

- Which patient and appointment characteristics are associated with higher no-show rates?

This may include factors such as age group, appointment type, appointment day, appointment time, booking lead time, distance to the clinic, and previous appointment history.

- Do patients with a history of previous no-shows have a higher likelihood of missing future appointments?

This question may help HealthConnect identify patterns of repeated missed appointments.

- Are appointment reminders associated with improved attendance and reduced no-show rates?

This will examine the potential relationship between reminders and appointment outcomes.

- Do no-show rates differ by reminder channel?

This will help investigate whether certain communication channels are associated with better attendance patterns.

- Are there particular days, times, or appointment types with consistently higher no-show rates?

This could help HealthConnect improve scheduling decisions and identify periods where intervention may be necessary.

5. Potential KPIs

At this stage, the KPIs are proposed for future analysis and are not yet being calculated.

KPI 1: No-Show Rate

Business Question: What proportion of scheduled appointments result in no-shows?

Definition: The percentage of relevant scheduled appointments that result in a No-Show outcome.

Why it is relevant: This is the primary KPI for measuring the scale of missed appointments and will help HealthConnect monitor whether attendance interventions are effective.

KPI 2: Appointment Attendance Rate

Business Question: What proportion of appointments are successfully attended?

Definition: The percentage of relevant scheduled appointments with an Attended outcome.

Why it is relevant: This KPI measures successful patient attendance and provides a direct view of appointment utilisation.

KPI 3: No-Show Rate by Patient and Appointment Segment

Business Question: Which patient and appointment characteristics are associated with higher no-show rates?

Definition: The no-show rate calculated across relevant segments such as age group, appointment type, appointment day, appointment time, booking lead time, and distance categories.

Why it is relevant: Comparing no-show rates across segments can help identify groups or appointment conditions associated with higher attendance risk.

KPI 4: Repeat No-Show Rate

Business Question: Do patients with previous no-show history have a higher likelihood of missing future appointments?

Definition: The no-show rate among patients with one or more previous no-shows compared with patients with no previous no-show history.

Why it is relevant: This KPI may help identify repeat patterns and support the development of targeted interventions for higher-risk patients.

KPI 5: Reminder Effectiveness Rate

Business Question: Are appointment reminders associated with improved attendance and reduced no-shows?

Definition: Comparison of attendance and no-show rates between appointments where a reminder was sent and appointments where no reminder was sent. This can also be analysed by the reminder channel.

Why it is relevant: This KPI can help HealthConnect evaluate whether its reminder process may be associated with better appointment attendance and whether certain reminder channels appear more effective.

6. Initial Analysis Approach

The project will follow a structured approach using SQL and Power BI.

Step 1: Load and Understand the Data

The dataset will first be imported into SQL for exploration.

The structure of the dataset will be reviewed by examining:

- Total number of records
- Number of variables
- Data types
- Unique values in important categorical fields
- Date ranges
- Appointment outcome categories

This step will ensure a clear understanding of the available data before further analysis.

Step 2: Perform Data Quality Checks

SQL queries will be used to investigate the following:

- Missing values
- Duplicate appointment records
- Unique appointment IDs
- Invalid or inconsistent values

- Logical relationships between variables
- Date consistency between booking and appointment dates
- Consistency between age and age group
- Consistency between appointment date and appointment day
- Relationships between reminder_sent and reminder_channel

Any necessary cleaning decisions will be documented before creating the final analysis dataset.

Step 3: Define the Analysis Population

A clear decision will be made regarding how cancelled appointments should be treated.

Since the main focus is appointment attendance and no-shows, one possible approach is to analyse no-show rates primarily among appointments with outcomes of 'Attended' or 'No-Show', while reporting cancellations separately.

This assumption will need to be documented because including cancellations in the denominator could change the calculated attendance and no-show rates.

Step 4: Conduct Exploratory Analysis

SQL will be used to calculate and compare appointment outcomes across relevant variables, including:

- Age group
- Gender
- Appointment type
- Appointment day
- Appointment time
- Booking lead days
- Previous appointments
- Previous no-shows
- Reminder status
- Reminder channel
- Distance to clinic
- Waiting time

The purpose of this stage will be to identify initial patterns and relationships that can be explored further.

Step 5: Prepare Data for Power BI

After data validation and initial SQL analysis, the cleaned dataset will be prepared for Power BI.

Additional calculated categories may be created where useful, such as:

- Booking lead time ranges
- Distance ranges
- Previous no-show categories
- Attendance indicator
- No-show indicator

These categories can make comparisons and visualisations easier to understand.

Step 6: Develop Power BI Analysis

Power BI will be used in the next stage to create interactive visualisations showing:

- Overall appointment outcomes
- Attendance and no-show rates
- No-show patterns by patient segment
- No-show patterns by appointment characteristics
- Reminder comparisons
- Time and scheduling patterns
- Potential high-risk appointment segments

The analysis will focus on identifying meaningful patterns that can support recommendations for improving appointment attendance.

7. Assumptions, Limitations, Risks and Dependencies

Assumptions

The analysis assumes that each record represents one appointment and that `appointment_id` is intended to uniquely identify each appointment.

It is also assumed that the appointment outcome recorded in the dataset is accurate and reflects the final status of each appointment.

Where `reminder_channel` is blank and `reminder_sent` is marked as No, the blank value will be treated as a logical absence of a reminder channel rather than missing data caused by an error.

Limitations

The dataset contains missing values for `distance_to_clinic_km` and `waiting_time_minutes`. These missing values may limit analysis involving these variables and will require an appropriate treatment strategy.

The dataset does not currently provide additional contextual information that may influence attendance, such as reason for cancellation, patient income, transport availability, weather conditions, clinic location, or detailed reminder timing.

Because the dataset is observational, relationships identified between variables should not automatically be interpreted as direct causation. For example, a difference in attendance between patients who received reminders and those who did not does not by itself prove that the reminder caused the difference.

Risks

Incorrect treatment of cancelled appointments could affect KPI calculations and conclusions about no-show rates.

Missing data could introduce bias if the missing values are not randomly distributed.

There is also a risk of drawing conclusions from associations without sufficient evidence of causation.

Patient information should be handled responsibly. Although the available patient IDs are identifiers rather than names, analysis and reporting should avoid exposing unnecessary patient-level information.

Dependencies

The next stages of the project depend on access to SQL and Power BI for data exploration and visualisation.

Further analysis also depends on establishing clear definitions for important metrics, particularly the denominator used for attendance and no-show rates and the treatment of cancelled appointments.

Project Summary

The Data Analytics track will address the problem of understanding appointment attendance and identifying patterns associated with patient no-shows at HealthConnect.

The main resources reviewed and used during Week 4 were the HealthConnect business scenario, project requirements, appointment dataset, dataset structure, and relevant variables. The analysis tools for the project will be SQL and Power BI.

The initial review found that the dataset contains 5,000 appointment records and 18 variables. No fully duplicated records were identified, and most data fields were complete. Missing values were identified in reminder_channel, distance_to_clinic_km, and waiting_time_minutes. The reminder channel values appear logically related to whether a reminder was sent, while the other missing values will require further consideration.

The proposed approach is to first validate and explore the data using SQL, establish clear definitions for attendance and no-show metrics, investigate patterns across patient and appointment characteristics, and then prepare the validated data for further analysis and visualisation in Power BI.

Key considerations include missing data, the treatment of cancelled appointments, the limitation of interpreting associations as causation, metric definitions, and responsible handling of patient-related information.

For Week 5, the proposed focus will be to begin detailed exploratory analysis using SQL, calculate the selected KPIs, investigate attendance and no-show patterns across relevant variables, and identify the most important areas for further Power BI analysis and visualisation.